Ashwin Guda

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EDUCATION

University of Florida, Herbert Wertheim College of Engineering Gainesville, FL
Bachelor of Science in Computer Science, GPA: 3.6 Aug 2024 – May 2028
Minor in Economics Dean's List

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Natural Language Processing, Intro to Software Engineering, Computer Organization, Discrete Structures

CERTIFICATIONS AND SKILLS

Certifications: Google Data Analytics Career Certificate
Skills: Python, SQL, C++, C#, Java, R, MATLAB, Git, AWS, Redis, Tableau, Snowflake, MS Excel, ROS, Linux, Claude Code

EXPERIENCE

Affinity Solutions New York City, NY
FI Data Analytics Intern Jun 2026 – Aug 2026

- Authored the technical specification for a new audience segmentation feature, defining the **SQL** aggregation logic and queries behind its insights dashboard, improving engineering workflows by **80%**
- Documented an end-to-end offer syndication pipeline spanning **SQL** storage tables, **Python** execution scripts, and validation rules improving onboarding by **50%**
- Audited transaction datamart tables produced by upstream **Airflow DAGs** against specification, reconciling schema and data mismatches before downstream use improving data accuracy by **25%**
- Wrote test cases and seeded pre-production **SQL** tables to validate **Python ETL** output across all user paths ahead of release

Mangomelon AI Remote
Data Analyst Intern Sep 2025 – Apr 2026

- Queried and processed **25M+** healthcare records across multiple NIS (National Inpatient Sample) tables (**8B+** total data values) using **SQL** and **Excel** and **AI** to analyze patient outcomes
- Analyzed NIS data using **MS Excel** and **SQL** to assess patient outcomes based on diagnoses, procedures, and comorbidities utilizing **AI-led** workflows

cSigma Finance New York City, NY
Summer Intern Jun 2025 – Aug 2025

- Analyzed company credit tapes to calculate liquidity and counter-party risk premiums utilizing **MS Excel**
- Wrote a white paper outlining ways to improve liquidity premium calculations accuracy within their credit model by **30%**

PROJECTS

Autonomous Wildfire Detection & Suppression Aug 2026 – Present

- Designing a two-stage **ML** detection framework fusing **RGB** and **thermal** imagery into a **6-channel** tensor, with temporal filtering and event-level evaluation splits to suppress false alarms
- Researching low-frequency acoustic flame suppression, modeling radiation efficiency to select waveguide confinement over free-field emission and designing a swept-frequency bench test
- Evaluating airframe and sensor tradeoffs for a patrol drone, sizing endurance, payload mass, and onboard **Jetson Orin NX** compute against fleet coverage requirements

Multithreaded Resale Market Intelligence Engine July 2026

- Built a concurrent **C++** data pipeline using a custom **thread pool** to fetch and aggregate sold listing data from **eBay's Browse API** across consumer goods categories
- Implemented **mutex-protected** aggregation to compute average resale margins, sales volume, and arbitrage opportunity scores across categories after platform fees

Campus Network Pathfinder April 2026

- Built a **C++** application modeling a weighted graph of campus locations to compute shortest routes and lowest-cost connection layouts
- Implemented **Dijkstra's algorithm** with a **min-heap priority queue** for efficient shortest-path
- Constructed a **minimum spanning tree** utilizing **Kruskal's** to create most efficient route layout for students